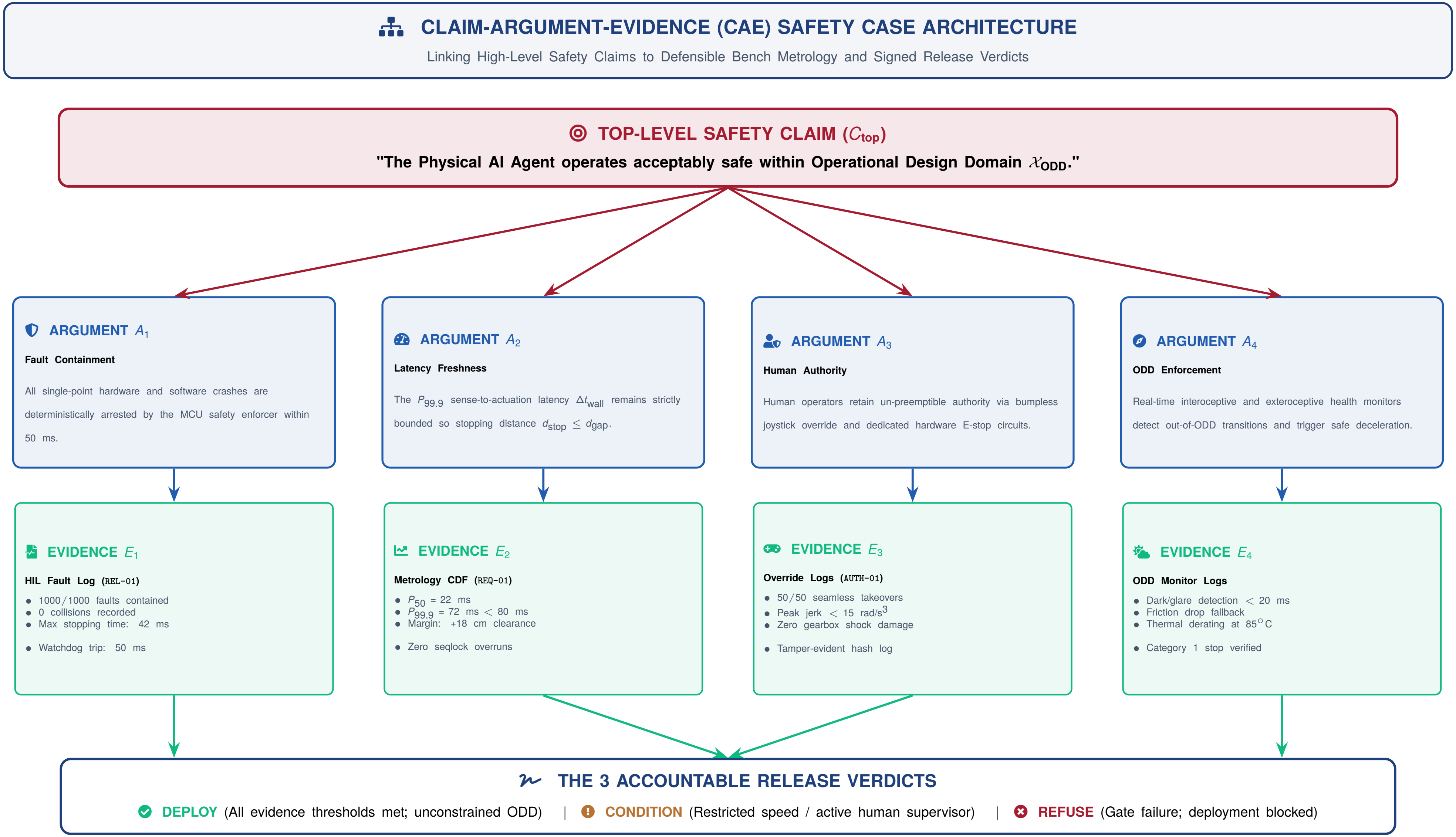
CLAIM-ARGUMENT-EVIDENCE (CAE) SAFETY CASE ARCHITECTURE
Linking High-Level Safety Claims to Defensible Bench Metrology and Signed Release Verdicts
TOP-LEVEL SAFETY CLAIM (C_{top})
"The Physical AI Agent operates acceptably safe within Operational Design Domain $\mathcal{X}_{ODD}$."
ARGUMENT A_1
Fault Containment
All single-point hardware and software crashes are deterministically arrested by the MCU safety enforcer within 50 ms.
ARGUMENT A_2
Latency Freshness
The $P_{99.9}$ sense-to-actuation latency Δt_{wall} remains strictly bounded so stopping distance $d_{stop} \leq d_{gap}$.
ARGUMENT A_3
Human Authority
Human operators retain un-preemptible authority via bumpless joystick override and dedicated hardware E-stop circuits.
ARGUMENT A_4
ODD Enforcement
Real-time interoceptive and exteroceptive health monitors detect out-of-ODD transitions and trigger safe deceleration.
EVIDENCE E_1
HIL Fault Log (REL-01)
1000/1000 faults contained
0 collisions recorded
Max stopping time: 42 ms
Watchdog trip: 50 ms
EVIDENCE E_2
Metrology CDF (REQ-01)
P_{50} = 22 ms
$P_{99.9}$ = 72 ms < 80 ms
Margin: +18 cm clearance
Zero seqlock overruns
EVIDENCE E_3
Override Logs (AUTH-01)
50/50 seamless takeovers
Peak jerk < 15 rad/s^3
Zero gearbox shock damage
Tamper-evident hash log
EVIDENCE E_4
ODD Monitor Logs
Dark/glare detection < 20 ms
Friction drop fallback
Thermal derating at 85°C
Category 1 stop verified
THE 3 ACCOUNTABLE RELEASE VERDICTS
DEPLOY (All evidence thresholds met; unconstrained ODD)
CONDITION (Restricted speed / active human supervisor)
REFUSE (Gate failure; deployment blocked)